LABSHEET-4

1.Create an XML document containing information about **five students**, including Student ID, Name, Course, Semester, Email, and Marks.

```
<?xml version="1.0" encoding="UTF-8"?>


<Students>


  <Student>
    <StudentID>101</StudentID>
    <Name>Varnika Tyagi</Name>
    <Course>BCA</Course>
    <Semester>3</Semester>
    <Email>varnika@gmail.com</Email>
    <Marks>85</Marks>
  </Student>


  <Student>
    <StudentID>102</StudentID>
    <Name>Rahul Sharma</Name>
    <Course>BCA</Course>
    <Semester>3</Semester>
    <Email>rahul@gmail.com</Email>
    <Marks>90</Marks>
  </Student>


  <Student>
```

```
  <StudentID>103</StudentID>
  <Name>Priya Verma</Name>
  <Course>BBA</Course>
  <Semester>2</Semester>
  <Email>priya@gmail.com</Email>
  <Marks>88</Marks>
</Student>

<Student>
  <StudentID>104</StudentID>
  <Name>Aman Kumar</Name>
  <Course>BCA</Course>
  <Semester>3</Semester>
  <Email>aman@gmail.com</Email>
  <Marks>78</Marks>
</Student>

<Student>
  <StudentID>105</StudentID>
  <Name>Neha Singh</Name>
  <Course>BBA</Course>
  <Semester>2</Semester>
  <Email>neha@gmail.com</Email>
  <Marks>92</Marks>
</Student>
```

</Students>

2.Create an XML document to store details of at least **five books**, including Book ID, Title, Author, Publisher, Price, and Category.

```
<?xml version="1.0" encoding="UTF-8"?>

<Books>

  <Book>
    <BookID>101</BookID>
    <Title>Wings of Fire</Title>
    <Author>A.P.J. Abdul Kalam</Author>
    <Publisher>Universities Press</Publisher>
    <Price>250</Price>
    <Category>Biography</Category>
  </Book>

  <Book>
    <BookID>102</BookID>
    <Title>Rich Dad Poor Dad</Title>
    <Author>Robert Kiyosaki</Author>
    <Publisher>Plata Publishing</Publisher>
    <Price>350</Price>
    <Category>Finance</Category>
  </Book>

  <Book>
```

```
  <BookID>103</BookID>
  <Title>Harry Potter</Title>
  <Author>J.K. Rowling</Author>
  <Publisher>Bloomsbury</Publisher>
  <Price>450</Price>
  <Category>Fantasy</Category>
</Book>

<Book>
  <BookID>104</BookID>
  <Title>The Alchemist</Title>
  <Author>Paulo Coelho</Author>
  <Publisher>HarperCollins</Publisher>
  <Price>299</Price>
  <Category>Fiction</Category>
</Book>

<Book>
  <BookID>105</BookID>
  <Title>Think and Grow Rich</Title>
  <Author>Napoleon Hill</Author>
  <Publisher>Rupa Publications</Publisher>
  <Price>200</Price>
  <Category>Self-Help</Category>
</Book>
```

</Books>

3.Create an HTML webpage containing a list of students and use JavaScript DOM methods such as getElementById(), getElementsByTagName(), and querySelector() to access and display the information.

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Information</title>
</head>
<body>

  <h2 id="heading">Student List</h2>

  <ul id="students">
    <li>Varnika Tyagi - BCA</li>
    <li>Rahul Sharma - BCA</li>
    <li>Priya Verma - BBA</li>
    <li>Aman Kumar - BCA</li>
    <li>Neha Singh - BBA</li>
  </ul>

  <h3>Student Information:</h3>

  <p id="result1"></p>
  <p id="result2"></p>
  <p id="result3"></p>
```

```
  <script>

    // Using getElementById()
    let heading = document.getElementById("heading");
    document.getElementById("result1").innerHTML =
      "Heading: " + heading.innerHTML;

    // Using getElementsByTagName()
    let students = document.getElementsByTagName("li");
    document.getElementById("result2").innerHTML =
      "Total Students: " + students.length;

    // Using querySelector()
    let firstStudent = document.querySelector("#students li");
    document.getElementById("result3").innerHTML =
      "First Student: " + firstStudent.innerHTML;

  </script>

</body>
</html>
```

4.Create an HTML page containing a heading, paragraph, and button. Use JavaScript DOM to **change the heading text, paragraph content, and background color** when the button is clicked.

```
<!DOCTYPE html>
<html>
<head>
```

```
  <title>DOM Manipulation</title>
</head>
<body>

  <h1 id="heading">Welcome to My Webpage</h1>

  <p id="para">This is the original paragraph.</p>

  <button onclick="changeContent()">Click Me</button>

  <script>

    function changeContent() {

      // Change heading text
      document.getElementById("heading").innerHTML =
        "Welcome to JavaScript DOM!";

      // Change paragraph content
      document.getElementById("para").innerHTML =
        "The paragraph content has been changed successfully.";

      // Change background color
      document.body.style.backgroundColor = "lightblue";

    }
```

```
  </script>


</body>

</html>
```

5.Create an XML file containing student records and use JavaScript/DOM concepts to **read and display the student information dynamically on an HTML webpage**.

```
<?xml version="1.0" encoding="UTF-8"?>


<Students>


  <Student>
    <ID>101</ID>
    <Name>Varnika Tyagi</Name>
    <Course>BCA</Course>
    <Semester>3</Semester>
    <Marks>85</Marks>
  </Student>


  <Student>
    <ID>102</ID>
    <Name>Rahul Sharma</Name>
    <Course>BCA</Course>
    <Semester>3</Semester>
    <Marks>90</Marks>
  </Student>
```

```
  <Student>
    <ID>103</ID>
    <Name>Priya Verma</Name>
    <Course>BBA</Course>
    <Semester>2</Semester>
    <Marks>88</Marks>
  </Student>

  <Student>
    <ID>104</ID>
    <Name>Aman Kumar</Name>
    <Course>BCA</Course>
    <Semester>3</Semester>
    <Marks>78</Marks>
  </Student>

  <Student>
    <ID>105</ID>
    <Name>Neha Singh</Name>
    <Course>BBA</Course>
    <Semester>2</Semester>
    <Marks>92</Marks>
  </Student>

</Students>
```

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Records</title>
</head>
<body>

  <h2>Student Information</h2>

  <button onclick="loadStudents()">Display Students</button>

  <table border="1" cellpadding="10">
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Course</th>
      <th>Semester</th>
      <th>Marks</th>
    </tr>

    <tbody id="studentData"></tbody>
  </table>

  <script>

    function loadStudents() {
```

```
let xhr = new XMLHttpRequest();

xhr.open("GET", "students.xml", true);

xhr.onload = function() {

  if (xhr.status === 200 || xhr.status === 0) {

    let xmlDoc = xhr.responseXML;

    let students = xmlDoc.getElementsByTagName("Student");

    let output = "";

    for (let i = 0; i < students.length; i++) {

      let id = students[i].getElementsByTagName("ID")[0].textContent;
      let name = students[i].getElementsByTagName("Name")[0].textContent;
      let course = students[i].getElementsByTagName("Course")[0].textContent;
      let semester = students[i].getElementsByTagName("Semester")[0].textContent;
      let marks = students[i].getElementsByTagName("Marks")[0].textContent;

      output += "<tr>";
      output += "<td>" + id + "</td>";
      output += "<td>" + name + "</td>";
```

```
                output += "<td>" + course + "</td>";
                output += "<td>" + semester + "</td>";
                output += "<td>" + marks + "</td>";
                output += "</tr>";
            }

            document.getElementById("studentData").innerHTML = output;
        }
      };

      xhr.send();
    }

  </script>

</body>
</html>
```